



Converting fractions to terminating decimals

Task A: Recognising terminating and non-terminating

1) Without a calculator, calculate the decimal value of the following fractions:

a) $\frac{7}{20} =$

d) $\frac{47}{200} =$

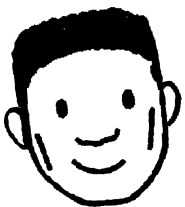
b) $\frac{5}{8} =$

e) $\frac{627}{50} =$

c) $\frac{3}{25} =$

f) $\frac{3}{16} =$

Lucas says



$\frac{33}{132}$ is not equivalent to a **terminating decimal**.

$$\frac{33}{132} = \frac{33}{2 \times 2 \times 3 \times 11}$$

The prime factors of the denominator are 2, 3 and 11.

2) Explain why $\frac{33}{132}$ is equivalent to a **terminating decimal**.

3) Jacob has discovered all these denominators will give a **terminating decimal**.

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256}$$

Can you explain why?

4) Circle all the **terminating decimals**.

$$\frac{2}{3}$$

$$\frac{3}{2}$$

$$\frac{4}{15}$$

$$\frac{12}{15}$$

$$\frac{7}{440}$$

$$\frac{11}{33}$$

$$\frac{6}{7}$$

$$\frac{11}{10}$$

$$\frac{11}{12}$$

$$\frac{7}{40}$$

$$\frac{110}{330}$$

Task B: Using a calculator

1) Using a calculator, match the fraction to its **terminating decimal**.

0.1294

13.444

0.39484

0.856

$$\frac{107}{125}$$

$$\frac{9871}{25\ 000}$$

$$\frac{647}{5000}$$

$$\frac{3361}{250}$$

2) Find the missing value(s) on the calculator displays.

